

ITA0606 – Machine Learning

Exp-11

Program for Credit Score Classification

Using Decision Tree Classifier

Step 1: Import required libraries

```
import pandas as pd
```

```
from sklearn.model_selection import train_test_split
```

```
from sklearn.tree import DecisionTreeClassifier
```

```
from sklearn.metrics import accuracy_score, confusion_matrix, classification_report
```

Step 2: Create the dataset

```
data = {
```

```
    'Age': [25, 35, 45, 23, 52, 40, 29, 60, 31, 48,  
            27, 55, 38, 42, 22, 50, 33, 58, 26, 46],
```

```
    'Income': [25000, 50000, 75000, 20000, 90000, 65000, 30000, 100000,  
               45000, 80000, 28000, 85000, 55000, 70000, 18000, 95000,  
               48000, 88000, 26000, 72000],
```

```
    'Loan_Amount': [10000, 20000, 15000, 12000, 10000, 18000, 25000, 8000,  
                    22000, 12000, 18000, 9000, 20000, 14000, 15000, 7000,  
                    23000, 6000, 20000, 11000],
```

```
    'Credit_History': [1, 1, 1, 0, 1, 1, 1, 1, 1, 1,  
                        0, 1, 1, 1, 0, 1, 1, 1, 0, 1],
```

```
    'Credit_Score': [
```

```

        'Good', 'Good', 'Good', 'Poor', 'Good',
        'Good', 'Average', 'Good', 'Average', 'Good',
        'Poor', 'Good', 'Average', 'Good', 'Poor',
        'Good', 'Average', 'Good', 'Poor', 'Good'
    ]
}

df = pd.DataFrame(data)

# Step 3: Display the dataset
print("Credit Score Dataset:")
print(df)

# Step 4: Separate features and target
X = df[['Age', 'Income', 'Loan_Amount', 'Credit_History']]
y = df['Credit_Score']

# Step 5: Split data into training and testing sets
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

print("\nTraining samples:", len(X_train))
print("Testing samples:", len(X_test))

# Step 6: Create Decision Tree Classifier
model = DecisionTreeClassifier(
    criterion='entropy',

```

```
    random_state=42
)

# Step 7: Train the model
model.fit(X_train, y_train)

# Step 8: Predict the test data
y_pred = model.predict(X_test)

# Step 9: Display predicted and actual values
print("\nPredicted Credit Scores:")
print(y_pred)

print("\nActual Credit Scores:")
print(y_test.values)

# Step 10: Calculate accuracy
accuracy = accuracy_score(y_test, y_pred)

print("\nAccuracy:", accuracy)
print("Accuracy Percentage:", accuracy * 100, "%")

# Step 11: Display confusion matrix
print("\nConfusion Matrix:")
print(confusion_matrix(y_test, y_pred))

# Step 12: Display classification report
print("\nClassification Report:")
```

```
print(classification_report(y_test, y_pred, zero_division=0))
```

OUTPUT

```
Credit Score Dataset:
```

	Age	Income	Loan_Amount	Credit_History	Credit_Score
0	25	25000	10000	1	Good
1	35	50000	20000	1	Good
...					

```
Training samples: 16
```

```
Testing samples: 4
```

```
Predicted Credit Scores:
```

```
['Good' 'Good' 'Average' 'Poor']
```

```
Actual Credit Scores:
```

```
['Good' 'Good' 'Average' 'Poor']
```

```
Accuracy: 1.0
```

```
Accuracy Percentage: 100.0 %
```